

Integrated Guide

Scientific Data Analysis v2.0

Skill backend and Scientific Workbench

Local-first product: contracts, guided workflows, run bundles, artifacts, jobs, previews, support, and dual validation

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Translation of the preserved private v2.0 guide at commit `b9f6176`. Original TeX SHA-256: `c154c0c20a8083a8e1ac8e315d6f2211f496a594a783200b1d78061db7a90701`. Private source: `skills/scientific-data-maintainer/.cache/active_references/guides/guia_scientific_data_analysis_v2_0.tex`.

Recorded results and closure steps are historical, not current verification. The original release declares installation complete only after Phase 11 post-synchronization validation of the installed skill and app. Historical commands are retained for traceability: use fresh output directories when reproducing them, without overwriting earlier evidence. The original remains unchanged privately.

Abstract

`scientific-data-analysis` v2.0 is the first release designed as an integrated product with Scientific Workbench. The skill remains the deterministic engine for scripts, JSON contracts, run bundles, artifacts, warnings, errors, provenance, and original preservation. Scientific Workbench is the macOS app for planning, guidance, execution, jobs, previews, support bundles, and confirmation controls.

Scientific execution remains local-first, with read-only originals or copied inputs. The original guide presents Ollama as the local AI route without a paid API, with OpenAI, Grok/xAI, and Gemini as optional cloud providers. The Codex bridge supports extended reasoning, code, and reporting; it does not replace deterministic capability execution.

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1. Executive overview

v2.0 brings together two previously separate pieces:

- `scientific-data-analysis`, with 59 public capabilities and app-readiness contracts.
- Scientific Workbench, a native macOS app for attaching data, reviewing plans, running workflows, and opening results.

The app supports planning and operation. Verifiable operational evidence lives in jobs, summaries, manifests, artifacts, logs, and run bundles; the release does not promise automatic scientific truth or mandatory cloud execution.

2. The skill

The deterministic backend:

- Registers public capabilities.
- Inspects and processes data, documents, FITS, notebooks, tables, and packages.
- Emits `summary.json`, `manifest.json`, `stdout`, `stderr`, and typed artifacts.
- Declares `PASS`, `WARNING`, `BLOCKED_CONTROLADO`, `FAIL`, or audit status `ROTO`.
- Preserves originals through copies or separate outputs.
- Blocks cleanly for absent optional backends or unsafe inputs.

3. Scientific Workbench

The product interface:

- Loads the installed skill registry.
- Accepts file or folder attachments.
- Uses the local planner, skill router, or AI to propose workflows.
- Executes capabilities in traceable runs.
- Shows jobs, progress, cancellation, retry, resume, and support bundles.
- Discovers PNG, PDF, CSV/TSV, Markdown, notebook, JSON, and log previews.
- Provides guided controls for column selection, optional panels, and visual confirmation.

4. Integration responsibilities

Layer	Responsibility
Skill	Reproducible scripts, envelopes, run bundles, artifacts, and provenance.
Scientific Workbench	UI, planning, command review, processes, artifact indexing, and job recovery.
Local/cloud AI	Explanation, planning, and assistance; not the sole scientific source of truth.
Codex bridge	Extended reasoning, code, reporting, and filesystem tasks under explicit control.

5. v2.0 JSON contract

The app accepts legacy, v1.8/v1.9, and v2.0 payloads. Main v2.0 fields:

Field	Use
<code>contract_version</code>	2.0 for new envelopes; earlier compatibility remains.
<code>status</code> / <code>app_status</code>	Normalized job/UI state.
<code>command</code>	Redacted command with <code>argv</code> , <code>cwd</code> , and provenance.
<code>inputs</code> / <code>outputs</code>	Declared inputs and outputs.
<code>typed_artifacts</code>	Primary artifact list for Results.
<code>warnings</code> / <code>errors</code>	Human-readable and parseable messages.
<code>next_actions</code>	Useful continuation actions.
<code>original_modified</code>	<code>false</code> , <code>true</code> , or explained <code>unknown</code> .
<code>app_hints</code>	Short summary, severity, suggested previews, and tags.
<code>job_metadata</code>	Retry, cancellation, exposure, and estimates.
<code>safety</code>	Read-only/copy policy, secret redaction, and confirmations.

6. Artifact types

Scientific Workbench recognizes frozen v1.8/v1.9 types and v2.0 additions:

```
summary_json, manifest_json, report_md, preview_png, table_csv,  
notebook_ipynb, log_txt, qa_report, handoff_bundle, unknown,  
fits_product, fits_visual, metadata_json, preview_pdf,  
edited_document, app_preview
```

Use `unknown` when an artifact cannot be typed accurately. Do not invent an ambiguous type to make the output appear more complete.

7. Run bundles

The preferred format remains:

```
run/  
  manifest.json  
  summary.json  
  stdout.txt  
  stderr.txt  
  command.txt  
  artifacts/  
  previews/  
  reports/  
  tables/  
  logs/  
  next_steps.md
```

The app can discover recursively but should prefer `typed_artifacts` and `manifest.json` when present.

8. Local-first, optional cloud, and Codex bridge

8.1. Local-first

The original guide recommends local Ollama: no paid API key is required, prompts/context remain on the local machine, and it supports planning, failure explanation, and result summaries.

8.2. Optional cloud

The guide lists OpenAI, Grok/xAI, and Gemini as optional providers for reasoning or writing. They can incur costs, require user keys, and do not execute capabilities themselves. Provider errors must be classified without leaking secrets. These are historical provider descriptions, not a current availability or pricing verification.

8.3. Codex bridge

Codex supports code, reports, refactors, and workspace reasoning. It does not replace the deterministic backend. Workflows must retain verifiable commands, outputs, artifacts, and logs.

9. Safety

The v2.0 integration requires:

- Read-only original inputs, with edits confined to copies or separate outputs.
- No API keys or secrets in logs, summaries, support bundles, or exportable prompts.
- Job records containing commands, statuses, warnings, errors, and next actions.
- Preflight and confirmation for optional/GUI backends, with clean blocking when prerequisites are absent.
- A release block for timeout, raw traceback, secret leakage, false success, or original modification.

10. Workflows recorded in Phase 8

Family	Status	Interpretation
Professional table	PASS	Parseable run bundle with artifacts.
Administrative document	WARNING	Useful intake with explicit limitations.
Inherited notebook	Controlled block	Stops interactive input or side effects before execution.
Corrupt container	WARNING	Recovers available content without false success.
Time series	WARNING	Visible date ambiguity.
Sensor/measurement	WARNING	Visible suspicious ranges.
FITS/RGB	WARNING	Successful RGB case; incomplete WCS warns.
Absent optional backend	Controlled block	Clean block with next actions.
DOCX roundtrip	PASS	Edits a confirmed copy, not the original.
Keynote/GUI	PASS	Preflight and confirmed export of a copy.

Controlled block denotes the original machine status `BLOCKED_CONTROLADO`.

11. Limits

- STILTS/TOPCAT, APT, Keynote, TEAREDUCE, IRAF, and iSTARMOD remain optional, expert, or legacy backends.
- Li I 6707.8 and SB2 tools are specialized, not general-purpose.
- GUI workflows need explicit confirmation and copied inputs.
- Cloud AI does not guarantee scientific accuracy.
- v2.0 is installed only after Phase 11 synchronizes the skill and validates the installed skill with Scientific Workbench.

12. Historical verification commands

The original workspace-specific paths are shown through aliases: `$PRIVATE_WORKSPACE` is the original editable workspace; `$DATANALYSIS_PYTHON` is its explicitly selected scientific interpreter. The external dual-gate script belongs to that workspace and is not supplied by this source publication. These are historical reproduction steps, not the current public installation procedure.

Main dual gate:

```
cd "$PRIVATE_WORKSPACE"
./run_v2_0_dual_gate.sh
```

Skill check:

```
cd "$PRIVATE_WORKSPACE/skill_work/scientific-data-analysis"
"$DATANALYSIS_PYTHON" scripts/sync_public_surface_docs.py --check
```

App check:

```
cd "$PRIVATE_WORKSPACE/ScientificWorkbench"
SKIP_UI_SMOKE=1 RUN_DOCUS_BENCHMARK=0 ./script/run_quality_gate.sh
```

13. Troubleshooting

Symptom	Next step
Absent optional backend	Accept <code>BLOCKED_CONTROLADO</code> ; inspect <code>preflight</code> and <code>next_actions</code> .
Output conflict	Use a fresh temporary folder or change <code>-outdir</code> .
Notebook with <code>input()</code>	Execute a copy with declared values or block before execution.
Missing preview	Inspect <code>manifest.json</code> , <code>typed_artifacts</code> , and <code>previews/</code> .
Cloud provider failure	Fall back to Ollama/local or the skill router; do not leak keys in logs.
Ambiguous result	Retain <code>WARNING</code> and explain the limit instead of silently reporting success.

14. Synchronization

Final synchronization belongs to Phase 11. Pre-sync gates must pass before copying the editable skill to the installed location. After copying, repeat checks on the installed skill and test Scientific Workbench against it.

Do not declare v2.0 installed when post-sync validation fails.

15. Historical decision

Phase 8 prepared the editable tree for joint documentation. Phase 9 produced this guide. Phase 10 validated the dual release candidate. Phase 11 installs the skill only if post-sync validation of the installed skill and Scientific Workbench passes without FAIL or ROTO.